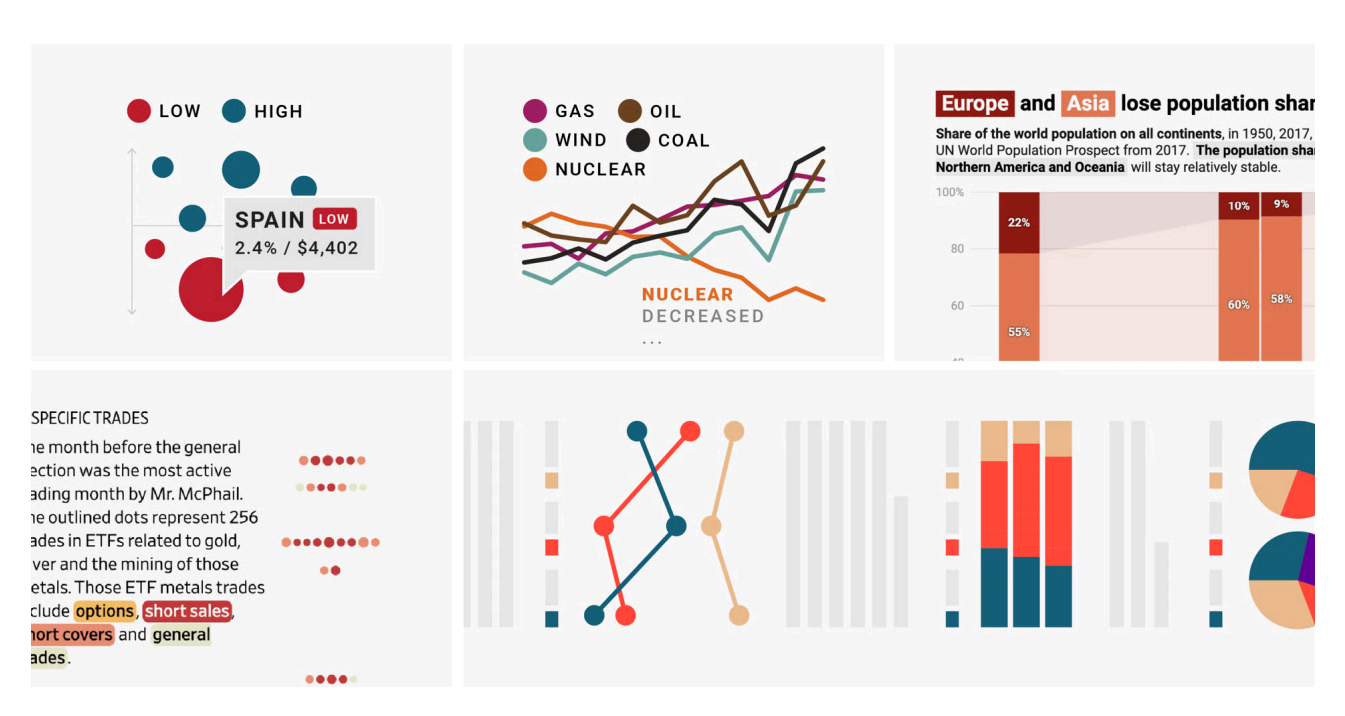
Color in data vis

14 min

October 11th, 2023

Remind readers of the colors in your data visualization

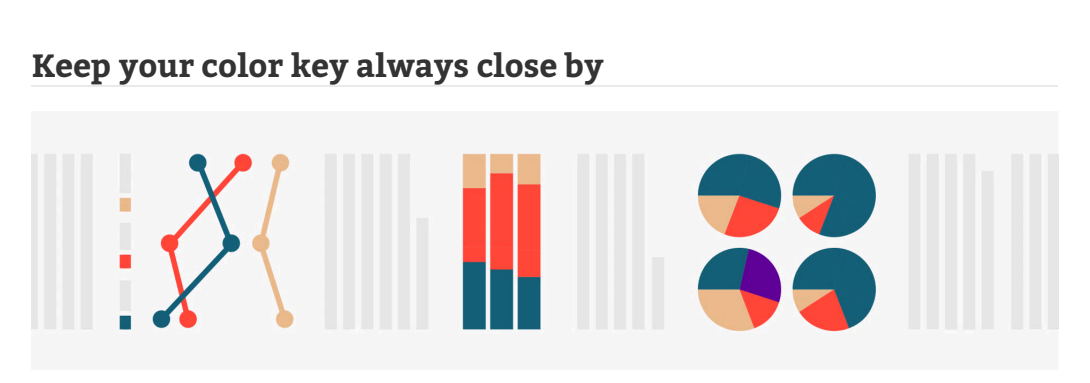
Lisa Charlotte Muth



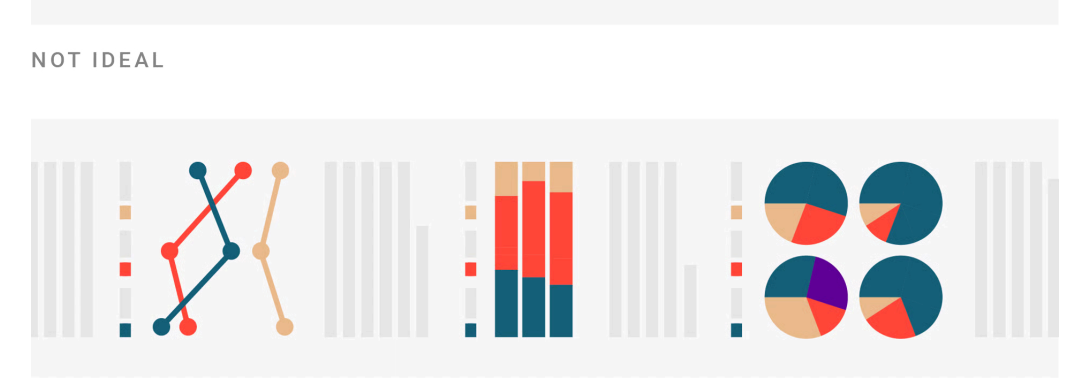
You **designed a useful (and fun!) color key for your data visualization** to explain to your readers what your colors mean? Great! Now you'll want to make sure people don't forget **those associations between colors and categories**. This article explores a few ways to do so:

1 Keep your color key always close by
2 Reuse your colors in annotations
3 Make your color key interactive
3 Reuse your colors in tooltips
5 Reuse your colors in text around your visualization

Keep your color key always close by



NOT IDEAL



BETTER

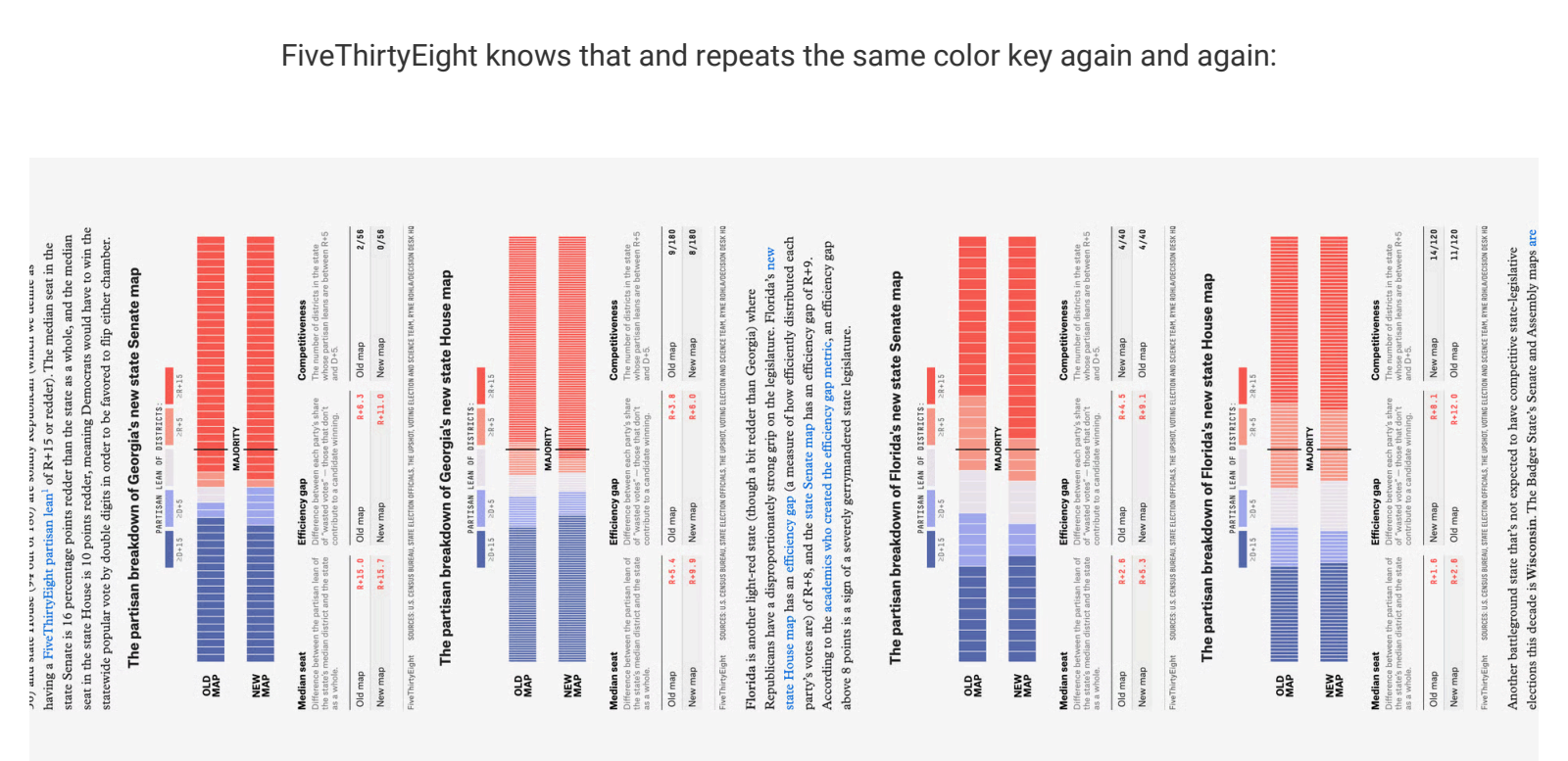
In a longer visualization, make sure the **color key is never far away.** In print, every visualization should be on the same page as its key. In digital formats, don't let readers scroll farther than roughly the height of their screen to see the key.

To achieve that in digital visualizations, make your color key **sticky**, so that as you scroll it stays visible on the top, bottom, or side:

The New York Times, 2023 The color legend and play button stick to the top of the screen as the reader scrolls through this tall data visualization.

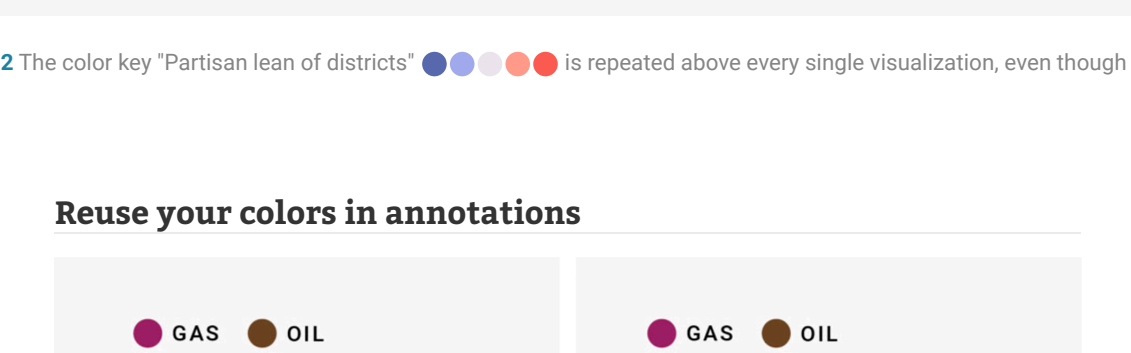
Another solution is to simply repeat your color key. It might be tempting to show the key only once — readers know the colors now, don't they? But if you pay attention, you'll notice that even you need to check the color key repeatedly when reading a data visualization for the first time. **Readers won't be annoyed by seeing the same color key three times,** but they will be annoyed if they can't quickly check what the colors are all about in the moment they want to.

FiveThirtyEight knows that and repeats the same color key again and again:



FiveThirtyEight, 2022 The color key "Partisan lean of districts" ●●●●● is repeated above every single visualization, even though they're fairly close together.

Reuse your colors in annotations



GOOD

BETTER

Annotations are often **already closer than the color key to your actual data points.** That makes them a neat and powerful tool for reinforcing your colors. When you repeat colors in an annotation, readers won't need to go searching for the color key and then traveling long distances between data and key with their eyes before understanding what your visualization is supposed to show.

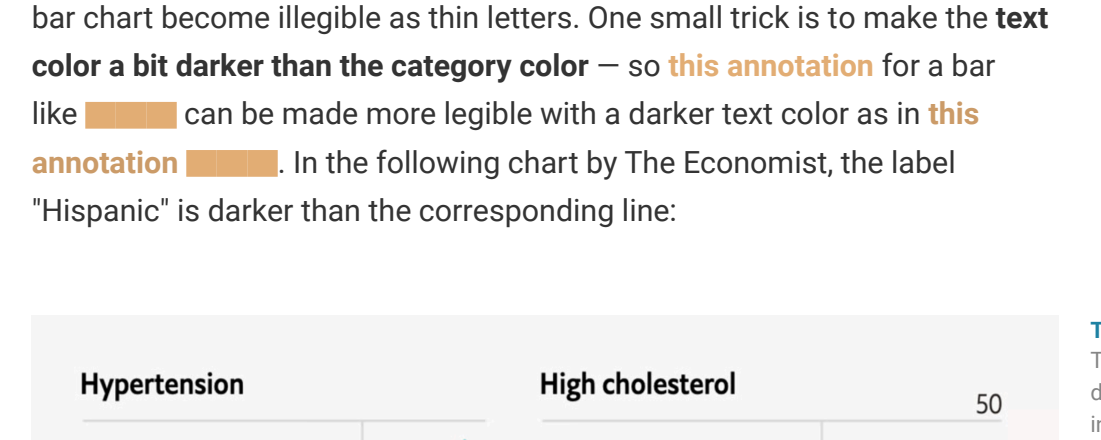
Annotations also **draw attention to certain data points**, so they become an additional way to lead a reader's eye to the categories you want them to see. (For other ways, read our article **Emphasize what you want readers to see with color.**)

For an effective annotation, make a statement about some specific data point and place the annotation right next to it. Then color the relevant part of your statement in the color of its corresponding category.

For example, in the annotation "Nuclear energy decreased its share since 2018," only the words "Nuclear" or "Nuclear energy" should be colored in orange. Orange stands for the **category** "Nuclear (energy)," not for the **fact** that the share of nuclear energy decreased.

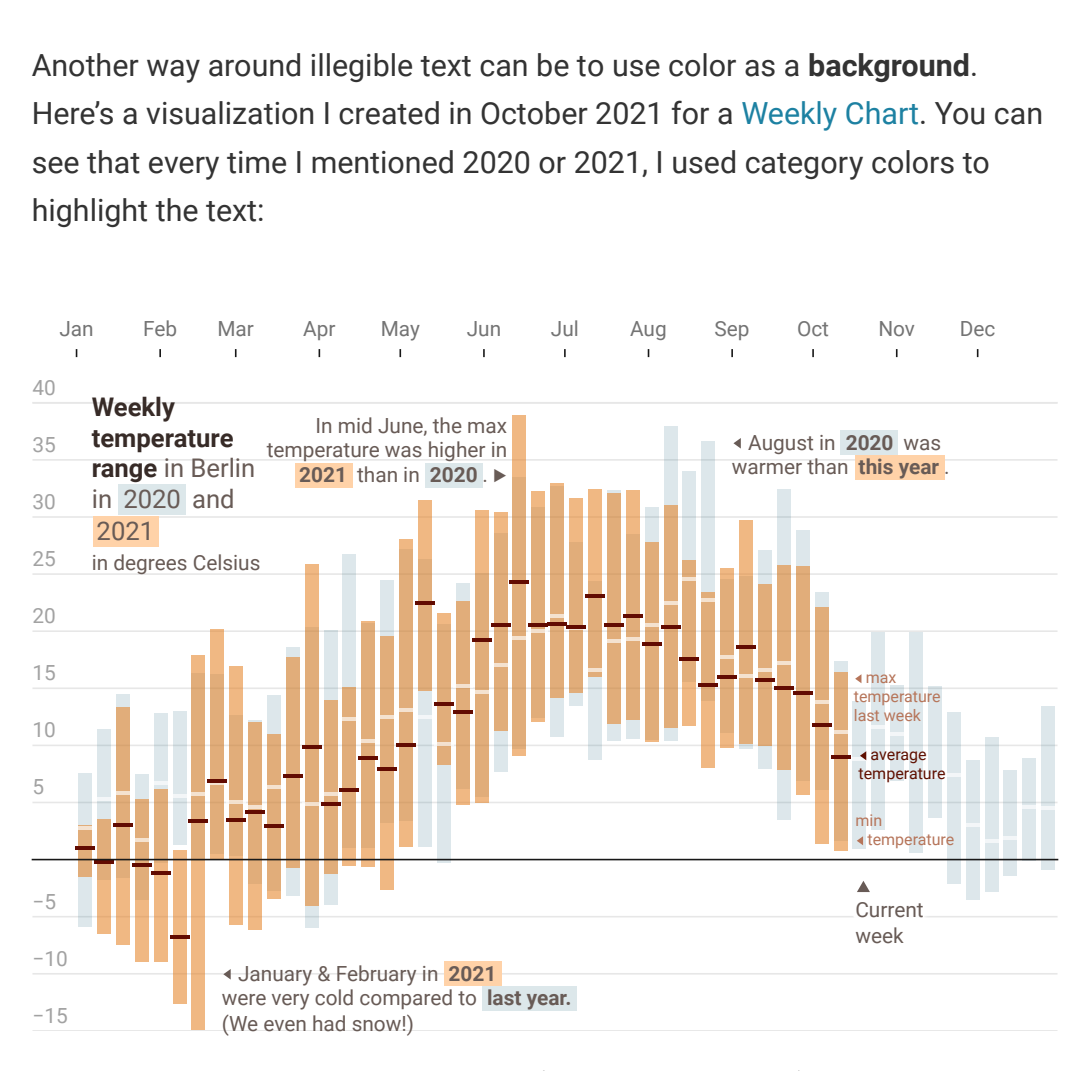
▶ How to color part of an annotation in Datawrapper

When you color annotation text to refer to a category of the same color, **the contrast between text color and background color should be high enough for reading.** Unfortunately, bright colors that work fairly well for a bar in a bar chart become illegible as thin letters. One small trick is to make the **text color a bit darker than the category color** — so this annotation for a bar like ▬ can be made more legible with a darker text color as in this annotation ▬. In the following chart by The Economist, the label "Hispanic" is darker than the corresponding line:



The Economist, 2023 The line showing the data for Hispanics is in a lighter gray ● than the word "Hispanic" ●

Another way around illegible text can be to use color as a **background.** Here's a visualization I created in October 2021 for a Weekly Chart. You can see that every time I mentioned 2020 or 2021, I used category colors to highlight the text:



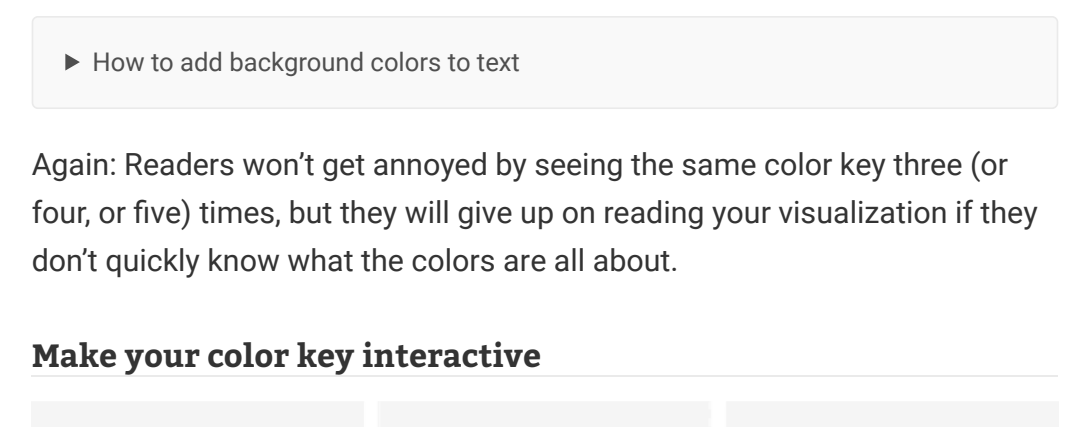
Weekly temperature range at a weather station in Berlin (Prenzlauer Berg, Bonsaiweg), Germany, in 2020 and 2021, in °C

Chart: Lisa Charlotte Muth, Datawrapper • Source: Weather Underground • Get the data • Created with Datawrapper

▶ How to add background colors to text

Again: Readers won't get annoyed by seeing the same color key three (or four, or five) times, but they will give up on reading your visualization if they don't quickly know what the colors are all about.

Make your color key interactive



NO INTERACTION

HOVER OVER DATA

HOVER OVER COLOR KEY

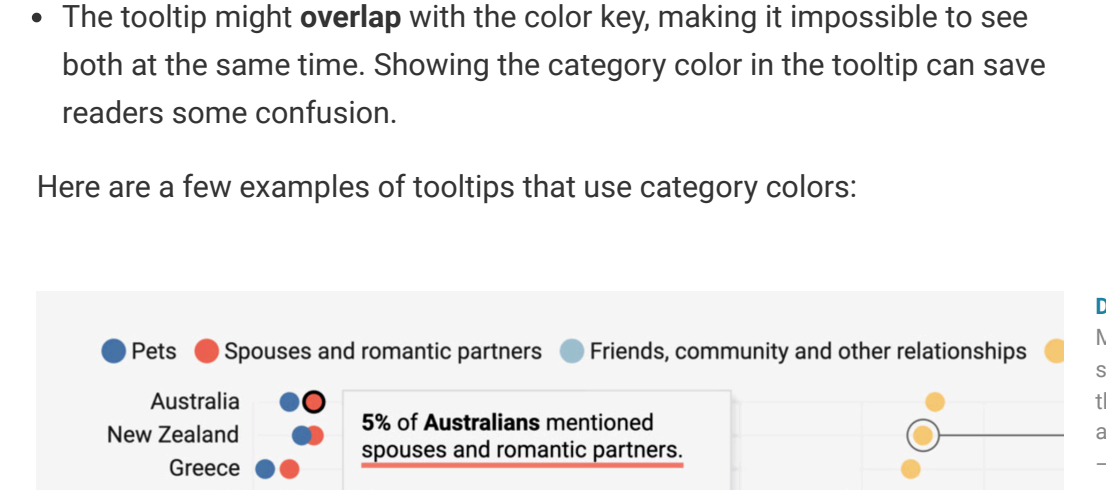
Especially if your visualization shows lots of different colors, interactivity can help readers to navigate them. This interactivity often goes both ways:

- If readers hover over a **data element** with a certain color, all colors but this one fade out in the color key.
- If readers hover over a color in the **color key**, all categories this one fade out in the visualization.

This is the default in many Datawrapper visualizations, be it stacked bar charts, line charts, pie charts, symbol maps, or choropleth maps. In the two kinds of maps, color keys for both quantitative and qualitative color scales can be made interactive, like so:

When readers hover over a point on a continuous color scale, the map will show all regions with approximately that value. It's fun! And helpful. Win-win!

Reuse your colors in tooltips



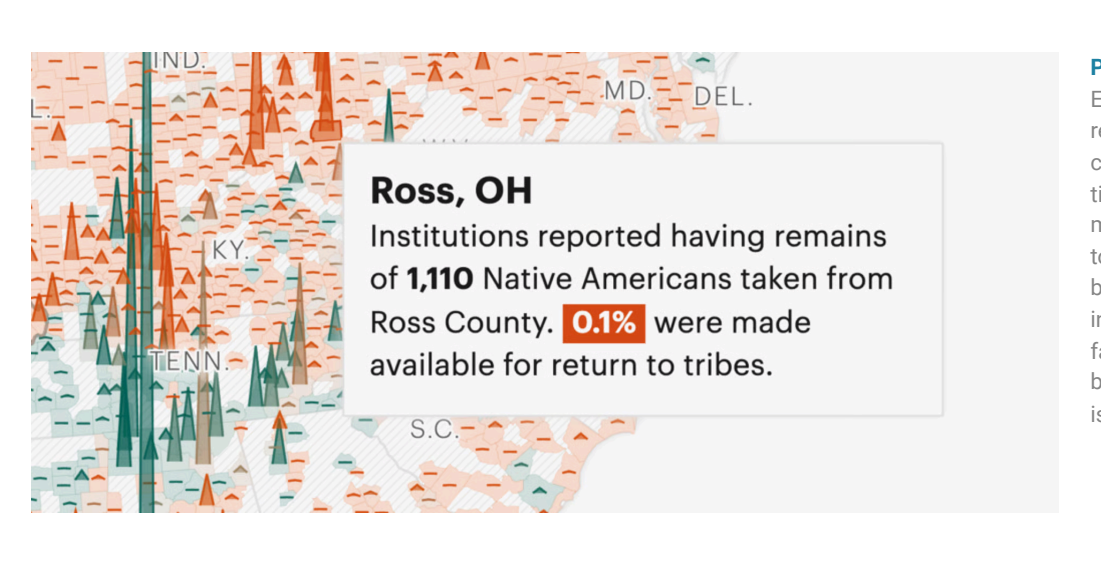
GOOD

BETTER

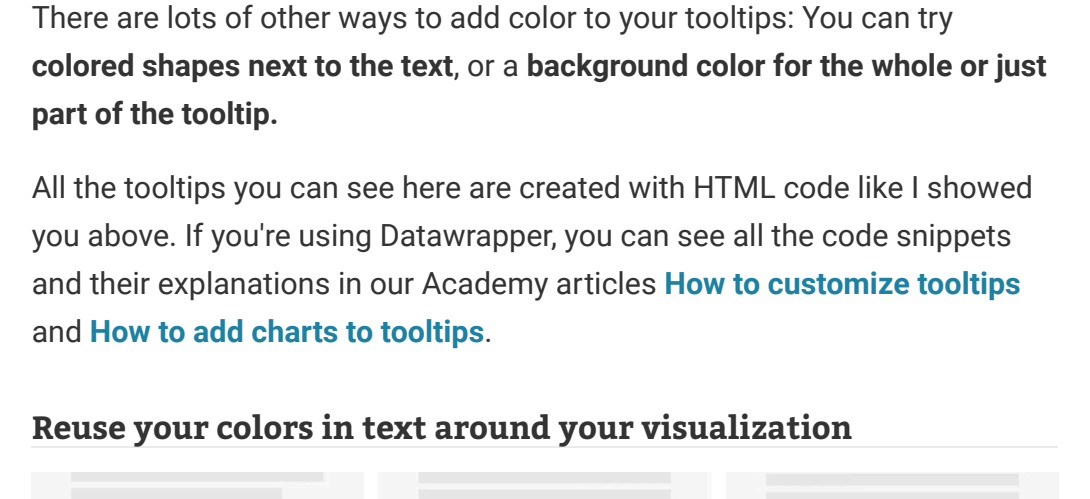
Tooltips, also called pop-ups, are another type of interactivity. Tooltips are great. They show the exact underlying data value(s) of each point in your visualization — and can use colors to remind readers of each point's category. There are a few reasons to use them that way:

- Readers might start interacting with your visualization before reading the color key. The tooltip can **teach** these readers which color stands for which category, while also **reminding** others who have read the key.
- Like annotations, tooltips are **very close** to the actual data points — which means that readers don't need to "eye-travel" between a potentially faraway color key and the data point.
- The tooltip might **overlap** with the color key, making it impossible to see both at the same time. Showing the category color in the tooltip can save readers some confusion.

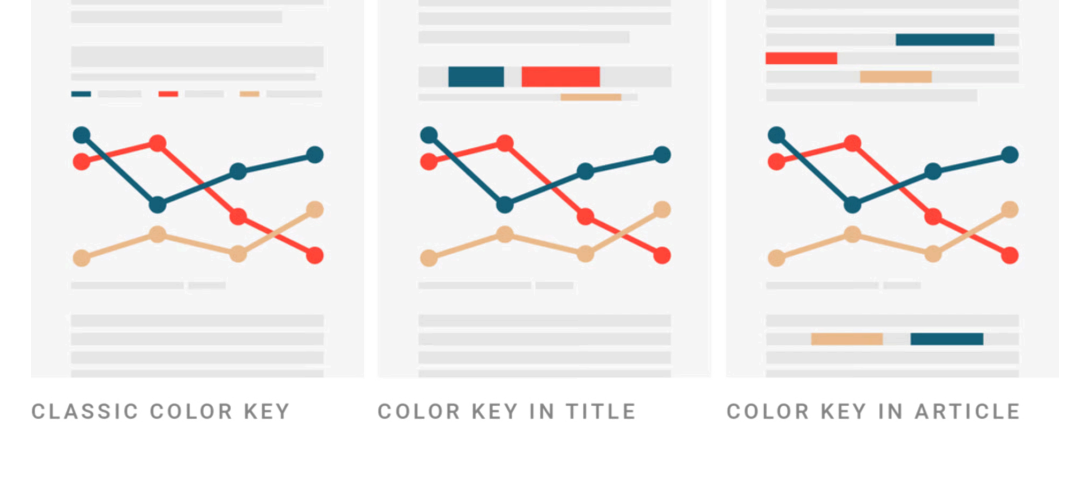
Here are a few examples of tooltips that use category colors:



Datawrapper, 2021 My coworker Rose stated the category in the tooltip – "spouses and romantic partners" – and underlined it with the category color ●. A nice touch: Using the same color to highlight the quote.



Datawrapper, 2020 For this map, I stated the category ("likely Biden") in the tooltip and colored it accordingly ●, but also used red and blue in the tooltip charts.

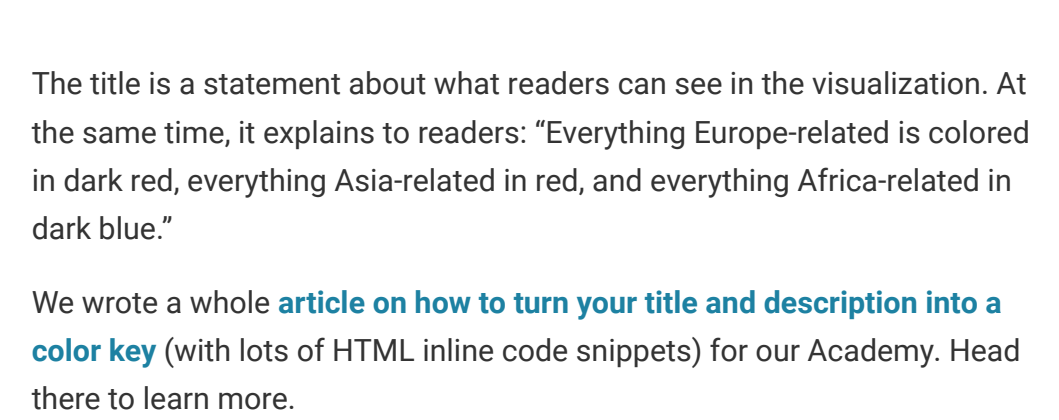

Ross, OH
Institutions reported having remains of 1,110 Native Americans taken from Ross County. 0% were made available for return to tribes.

ProPublica, 2023 Especially if the regions that readers can hover over are very small, it's crucial to make very clear in the tooltip which region is being highlighted and into which category it falls. In this case, a background color ● is used.

There are lots of other ways to add color to your tooltips: You can try **colored shapes next to the text**, or a **background color for the whole or just part of the tooltip.**

All the tooltips you can see here are created with HTML code like I showed you above. If you're using Datawrapper, you can see all the code snippets and their explanations in our Academy articles **How to customize tooltips** and **How to add charts to tooltips.**

Reuse your colors in text around your visualization



CLASSIC COLOR KEY

COLOR KEY IN TITLE

COLOR KEY IN ARTICLE

Your visualization and the text around it can be linked with colors. There are two ways to do so: Use the category colors in the **title and description**, or use the colors in the **article surrounding the visualization.**

Here's an example of the first approach:

Europe and **Asia** lose population share, **Africa** gains

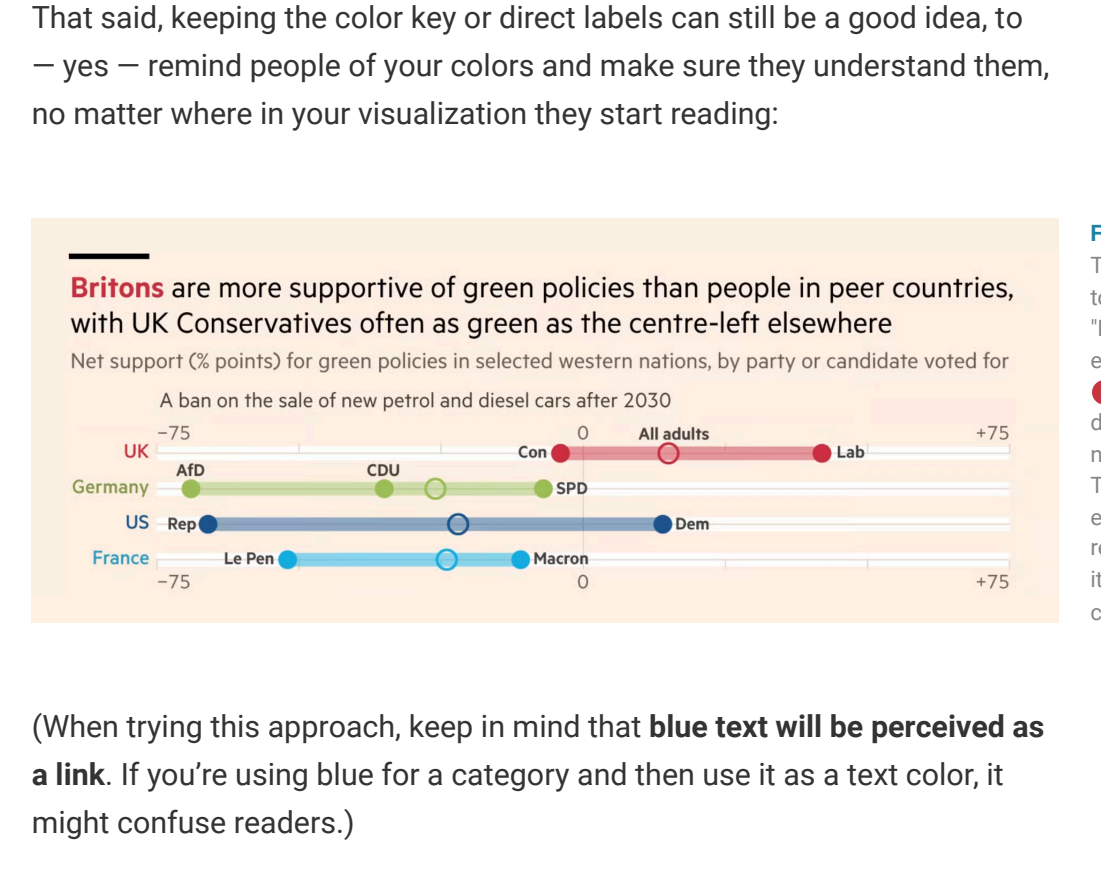
Share of the world population on all continents, in 1950, 2017, 2030, 2050 and 2100, according to the UN World Population Prospect from 2017. **The population share in Latin America and the Caribbean, Northern America and Oceania** will stay relatively stable.

Chart: Lisa Charlotte Muth, Datawrapper • Source: UN World Population Prospects, 2017

The title is a statement about what readers can see in the visualization. At the same time, it explains to readers: "Everything Europe-related is colored in dark red, everything Asia-related in red, and everything Africa-related in dark blue."

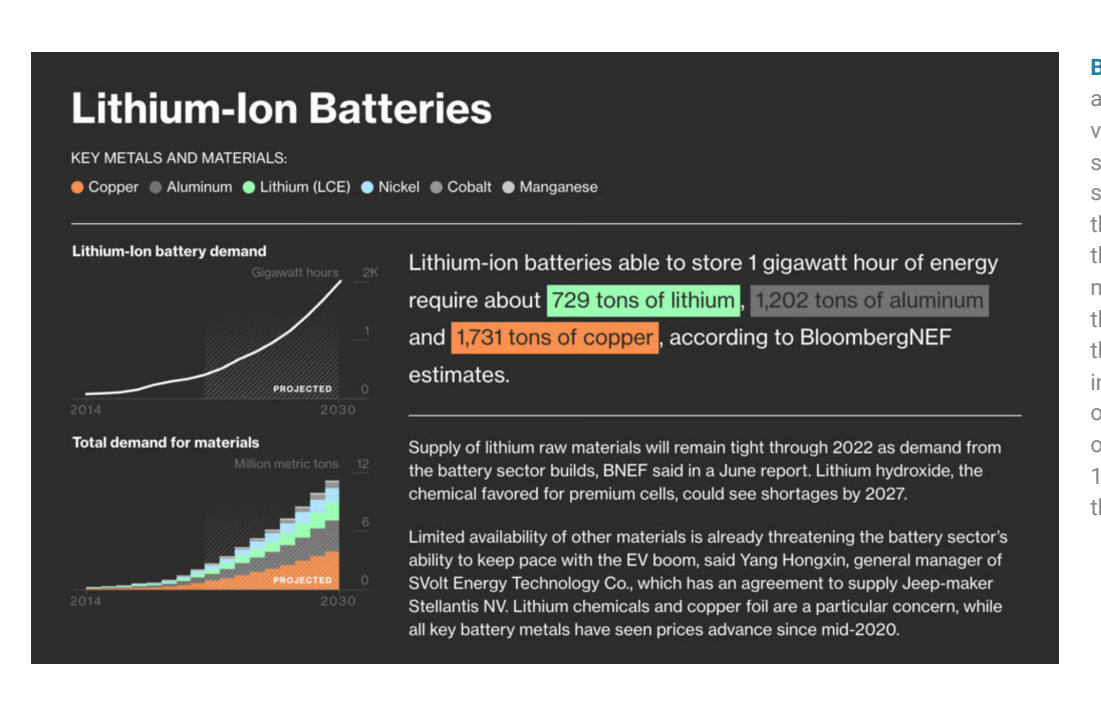
We wrote a whole **article on how to turn your title and description into a color key** (with lots of HTML inline code snippets) for our Academy. Head there to learn more.

Having the color key in the title means **you don't need an additional, "classic" color key** — it saves space and can make the visualization feel simpler and easier to understand:


64 Years of Moon Crashes
Seven space programs and one private company have made hard landings on the moon: the Soviet Union, United States, Japan, European Space Agency, India, China, Israel and ispace. Crash locations are shown on the map below.

The New York Times, 2023 Instead of adding a color key, this article lists and colors countries directly above the map. Note that the type of font changes from a serif to a sans-serif for the countries — it's the same typeface that's also used for the labels on the map.

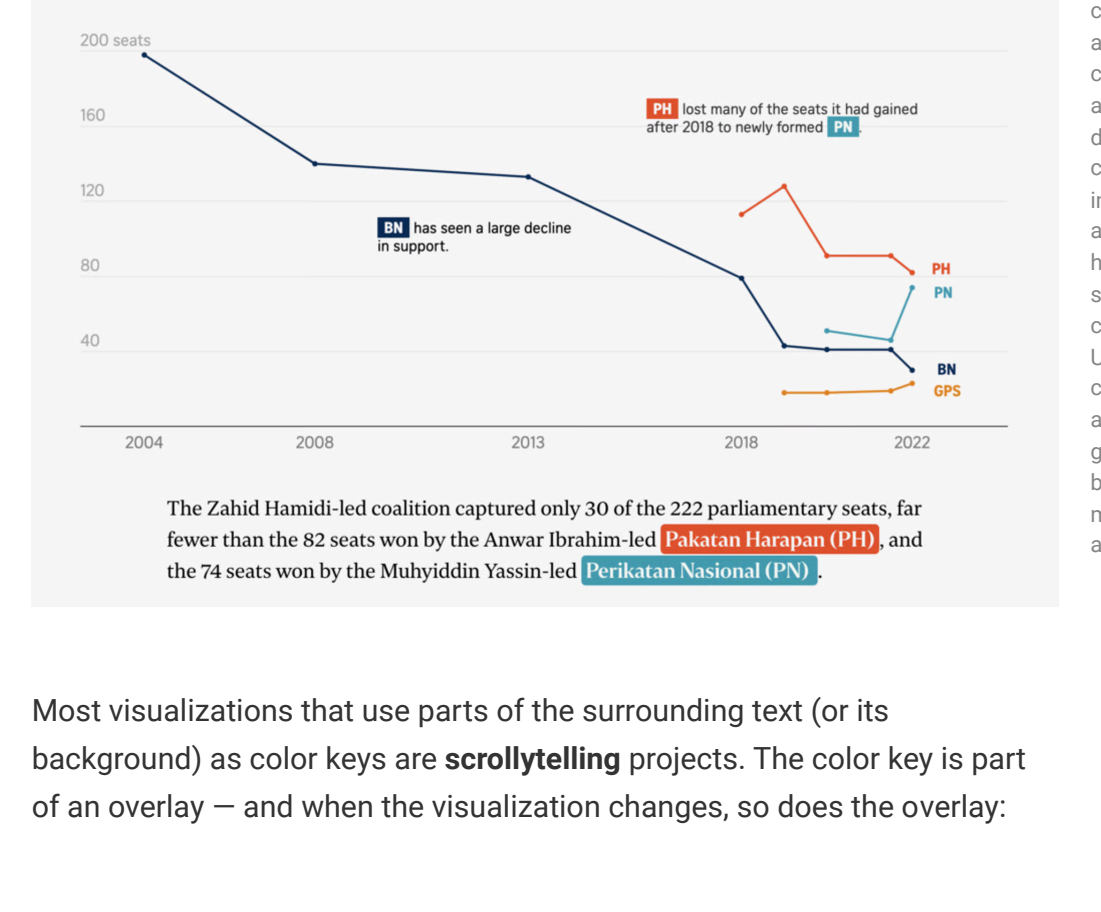
That said, keeping the color key or direct labels can still be a good idea, to — yes — remind people of your colors and make sure they understand them, no matter where in your visualization they start reading:



Financial Times, 2023 The title makes clear to readers that "Britons" can be expected to be in red ●, but the chart doesn't omit its normal label ("UK"). This double explanation is redundant, but makes it easier to read the chart.

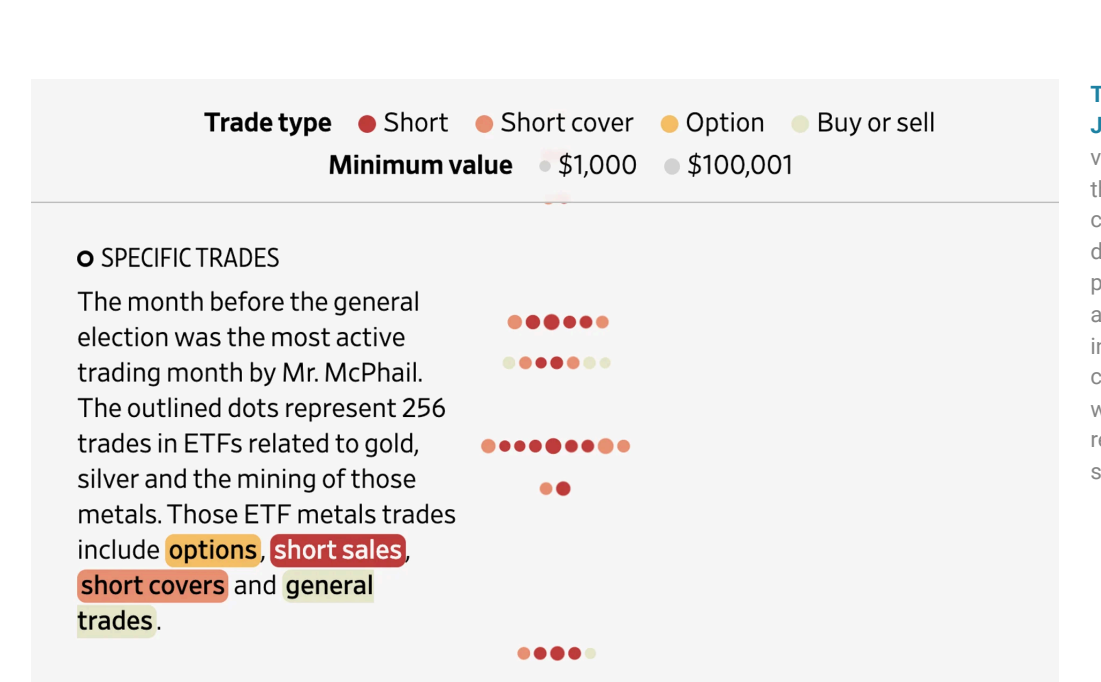
(When trying this approach, keep in mind that **blue text will be perceived as a link**. If you're using blue for a category and then use it as a text color, it might confuse readers.)

Including a color key in the surrounding article — and hence creating a deeper integration between article and visualization — seems to be less common. Here are two attempts, but both keep a traditional color key as well:



Lithium-ion batteries able to store 1 gigawatt hour of energy require about 729 tons of lithium, 1,202 tons of aluminum and 1,731 tons of copper, according to BloombergNEF estimates.

Bloomberg, 2021 The actual color key uses very small color swatches and a smaller font size than the text to the right of the visualization. For most readers, this has the effect of making the background colors in the text ("729 tons of lithium, 1,202 tons of aluminum and 1,731 tons of copper") the primary color key.



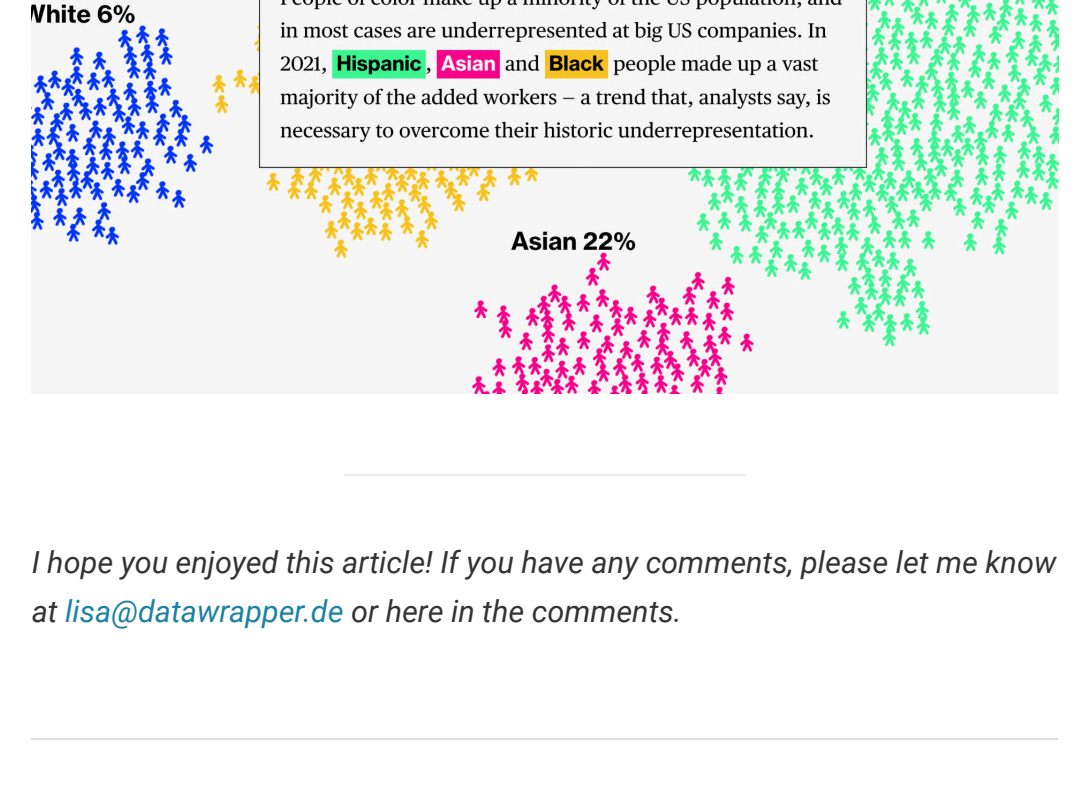
On a night of uncertainty, one thing was clear. Barisan Nasional (BN) had its worst electoral performance in decades.

The Zahid Hamidi-led coalition captured only 30 of the 222 parliamentary seats, far fewer than the 82 seats won by the Anwar Ibrahim-led Pakatan Harapan (PH), and the 74 seats won by the Muhyiddin Yassin-led Perikatan Nasional (PN).

The Straits Times, 2022 This article gives readers three color keys: in the article surrounding the chart, as part of the annotation, and as direct labels. The color keys feel more important than the actual line chart here; thicker, more self-confident lines could have helped. Underlining the categories in the article instead of giving them a background color might have been another solution.

Most visualizations that use parts of the surrounding text (or its background) as color keys are **scrollytelling** projects. The color key is part of an overlay — and when the visualization changes, so does the overlay:

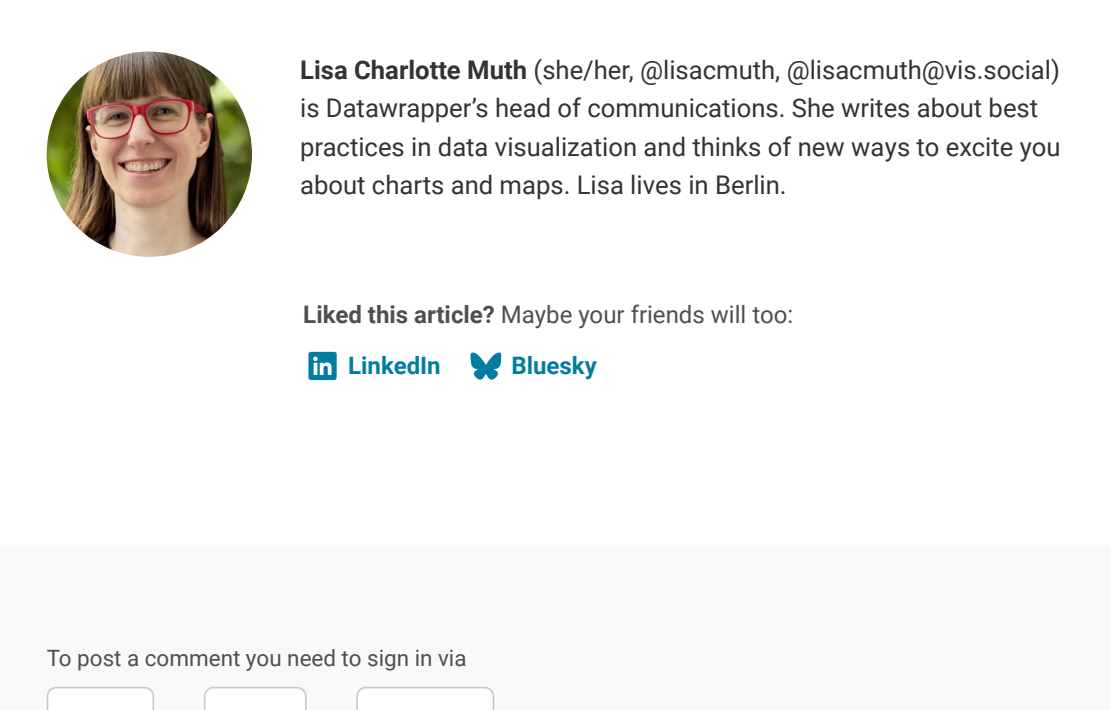
Bloomberg, 2021 Including the color key in the article is a common tactic for scrollytelling like this one.



SPECIFIC TRADES

The month before the general election was the most active trading month by Mr. McPhail. The outlined dots represent 256 trades in ETFs related to gold, silver and the mining of those metals. Those ETF metals trades include options, short sales, short covers and general trades.

The Wall Street Journal, 2022 Another visualization where the text functions as a color key that's designed more prominently than the actual key. To increase the color contrast, white text was chosen for the red category "short sales."



People of color make up a minority of the US population, and in most cases are underrepresented at big US companies. In 2021, Hispanic, Asian and Black people made up a vast majority of the added workers – a trend that, analysts say, is necessary to overcome their historic underrepresentation.

Bloomberg, 2023 Even if categories are directly labeled like here, an additional color key within the text can be useful.

I hope you enjoyed this article! If you have any comments, please let me know at lisa@datawrapper.de or here in the comments.

Lisa Charlotte Muth (she/her, @lisacmuth, @lisacmuth@vis.social) is Datawrapper's head of communications. She writes about best practices in data visualization and thinks of new ways to excite you about charts and maps. Lisa lives in Berlin.

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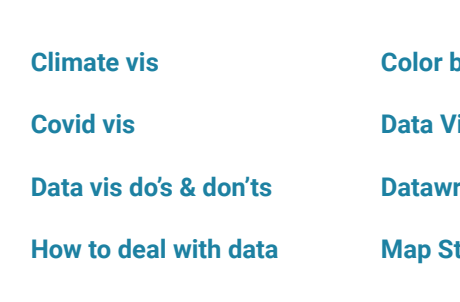

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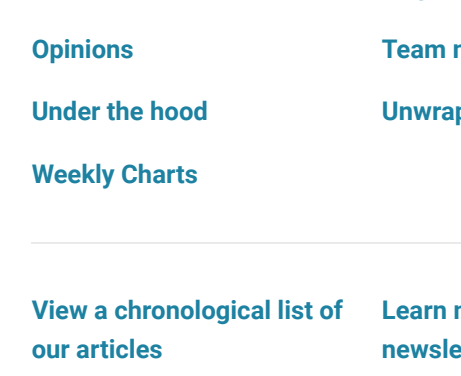
December 11th, 2024 by Lisa Charlotte Muth



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How to design a useful (and fun!) color key for your data visualization

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